

Implementing LISP09 on a 6809 board

 flexonsbd.blogspot.com/2019/04/6809lisp09.html

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I implemented LISP09 on a custom-made 6809 circuit board.

My attempt to implement GAME80 on the SBC8080 was unsuccessful and has been put on hold, so

I tried implementing LISP09 on my own 6809 board. LISP09 was created by Teruo Serizawa and published in "Transistor Technology 6809 All-Mighty Special Edition," which I borrowed from the library. The assembly listing was 23 pages long, but I managed to input it. This LISP09 was created as software for FLEX09, so it worked fine just by inputting it. It starts by typing LISP09 on FLEX09. All the sample programs worked, so I don't think there were any input errors.

```
M6809 MONITOR Ver6.1
>FLEX

6809 FLEX V3.01

Time : 2019/04/20 12:11:22

+++LISP09
-----
LISP-09 Interpreter version 2.08 1983.10.07
  Copyright (C) 1982 by Kogakuin University
-----
# of free cells : 7680
atom area, used : 1907
User stack area : 6285
System stack area: 2853
```

Startup screen and object list

```
COM3:38400bps - Tera Term VT
ファイル(F) 編集(E) 設定(S) コントロール(C) ウィンドウ(W) ヘルプ(H)

:(OPENR 'QUICK-SORT.TXT)
NIL
:(DE QUICK-SORT (L FN X)
:  (COND ((NULL L) L)
:      (T (SETQ X (POP L))
:          (LOOP (LEFT RIGHT)
:                (COND ((NULL L)
:                      (RETURN (NCONC (QUICK-SORT LEFT FN)
:                                     (CONS X
:                                           (QUICK-SORT RIGHT FN))))))
:      ((FN (CAR L) X)
:          (PUSH (POP L) LEFT))
:          (PUSH (POP L) RIGHT)
:      )
:  )
QUICK-SORT
:(SETQ TEST-DATA '(6809 -8080 68000 -80 6502 -8086))
(6809 -8080 68000 -80 6502 -8086)
:
:(QUICK-SORT TEST-DATA 'GREATERP)
(68000 6809 6502 -80 -8080 -8086)
:
```

The quicksort program shown in the example

It's been decades since I last touched LISP, and I don't remember anything, so I'll have to pull out Professor Kurokawa's "Introduction to LISP" from my bookshelf and start learning from the very beginning. [Added April 27th] I found one error in the input list, but it didn't affect the assembly result. I don't think there are any more errors, so I'll finish checking. It runs on FLEX9, but I think it might be possible to port it if the source code is available, so I'll upload the source code and assembly list to [OneDrive](#) . (Since it was published in a magazine , I

think it should be OK to make it public for personal use , but what do you think? Incidentally, Professor Serizawa seems to have published it in a paper as well .)